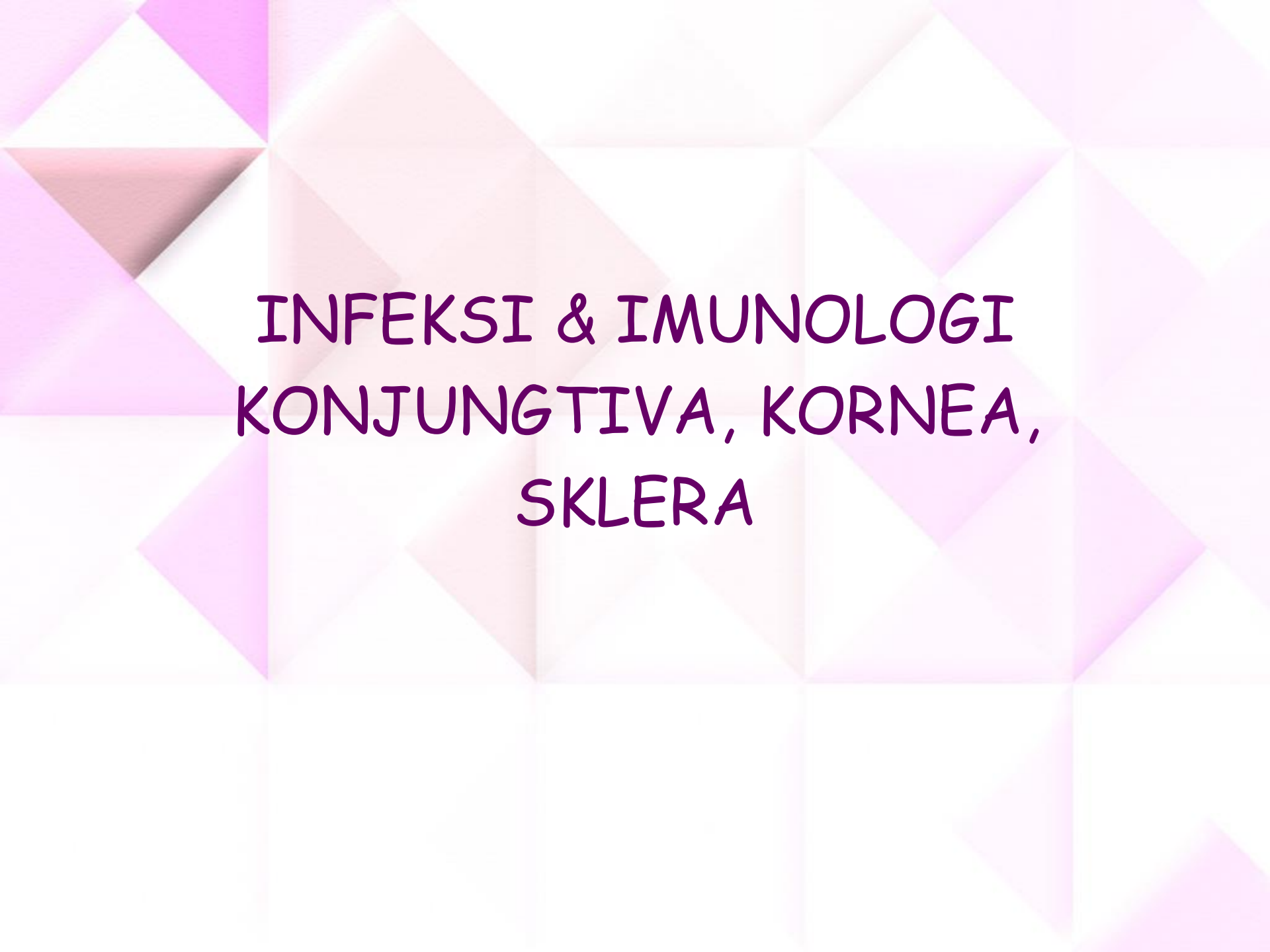




INFEKSI & IMUNOLOGI KONJUNGTIVA, KORNEA, SKLERA

KONJUNTIVA

ANATOMI

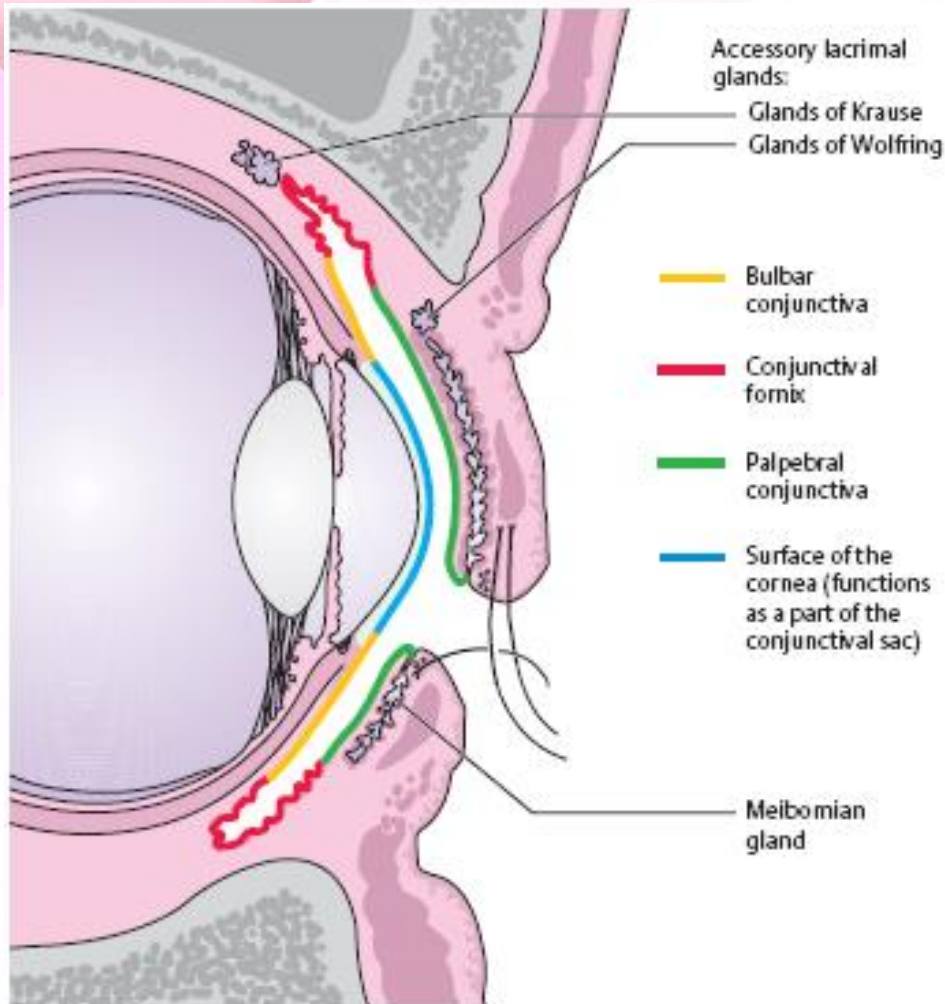


Fig. 4.1 The conjunctiva consists of the bulbar conjunctiva, the conjunctival fornices, and the palpebral conjunctiva. The surface of the cornea functions as the floor of the conjunctival sac.

CONJUNCTIVA:

- KONJUNGTTIVA BULBI → cover anterior sclera
- KONJUNGTTIVA PALPEBRA → from margin mucocutaneous junction-tarsal plate
- KONJUNGTTIVA FORNIKS

SACCUS CONJ. :

- GLOBE MOVEMENT
- ARTICULATING LAYER
- PROTECTIONI

ANATOMI

HISTOLOGIS

- EPITEL KONJUNGATIVA → sel kuboid di tarsus, kolumnar di forniks, dan skuamous di bola mata
- STROMA / SUBST. PROPRIA → jaringan ikat longgar dan vaskularisasi. Dipisahkan dengan membran basalis.

KELENJAR

- SEKRESI MUSIN → sel goblet, crypts of Henle, kelenjar Manz
- KELENJAR LAKRIMAL ASESORIA → kel. Krause dan Wolfring

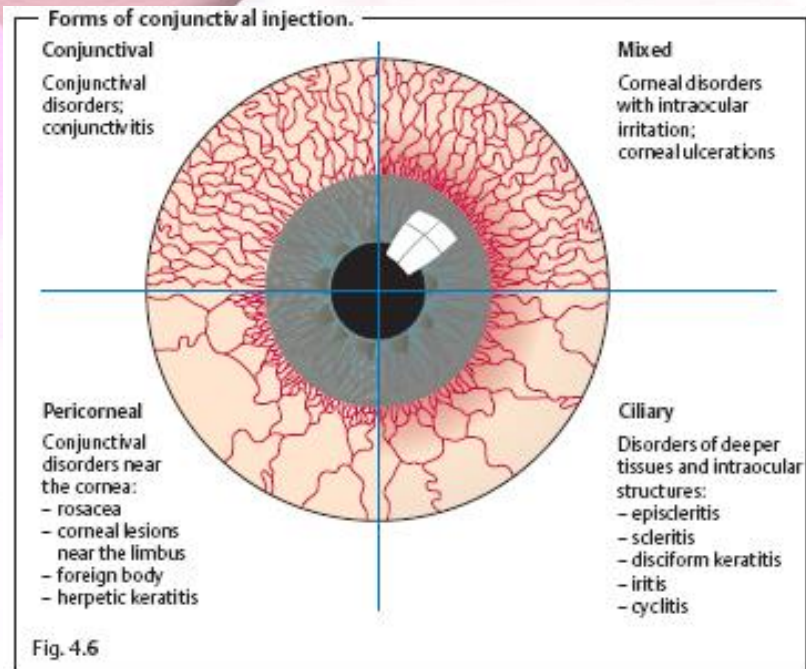
KONJUNGTIVITIS

Radang pada konjungtiva

SIGNS & SYMPTOMS :

1. KONJUNGTIVAL HIPEREMI
2. KONJUNGTIVAL EDEMA + PALPEBRA EDEMA
3. PAIN
4. DISCHARGE
5. OTHER

KONJUNGTIVAL HIPEREMI

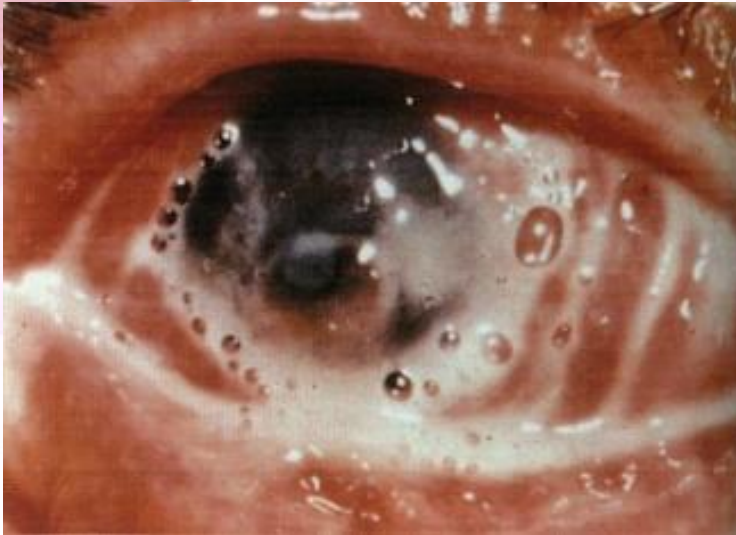


→ Gambaran >> di forniks

KONJUNGTTIVA + PALPEBRA EDEMA



DISCHARGE



- WATERY → infeksi virus
- MUCOID → vernal dan keratokonjungtivitis sicca
- PURULENTA → infeksi bakteri akut
- MUCOPURULENTA → infeksi bakteri ringan dan clamidia

LAIN-LAIN



SUBKONJUNTIVAL
HAEMMORRHAGE



SCARRING



REAKSI
FOLLICULAR

LAIN-LAIN



REAKSI
PAPILER



LIMBAL NODULE
(TRANTAS DOTS)



PSEUDOMEMBRAN

DIFFERENTIAL DIAGNOSA

Symptom or finding	Bacterial conjunctivitis	Chlamydial conjunctivitis	Viral conjunctivitis	Allergic conjunctivitis	Toxic conjunctivitis
Itching	–	–	–	++	–
Hyperemia (reddened eye)	++	+	+	+	+
Bleeding	+	–	+	–	–
Discharge	Purulent; yellow crusts	Mucopurulent	Watery	Ropy white, viscous	–
Chemosis	++	–	±	++	±
Lacrimation (epiphora)	+	+	++	+	+
Follicles	–	++	+	+	+
Papillae	+	±	–	+	–
Pseudomembranes, membranes	±	–	±	–	–
Swollen lymph nodes	+	+	++	–	–
Pannus formation	–	+	–	–	±

TERAPI

BACTERIAL CONJUNCTIVITIS

- Antibiotic drops
 - fusidic acid q.i.d 48 jam lalu b.d
 - chloramphenicol (broad spec.) every 1-2 jam
 - ciprofloxacin, ofloxacin, gentamycin, tobramycin
- Antibiotic ointments
 - higher concentration for longer period
 - chloramphenicol, gentamycin, tetracyclin

VIRAL CONJUNCTIVITIS

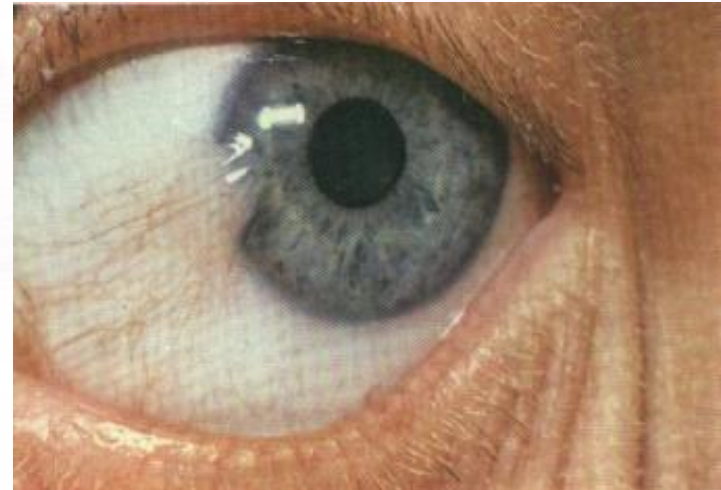
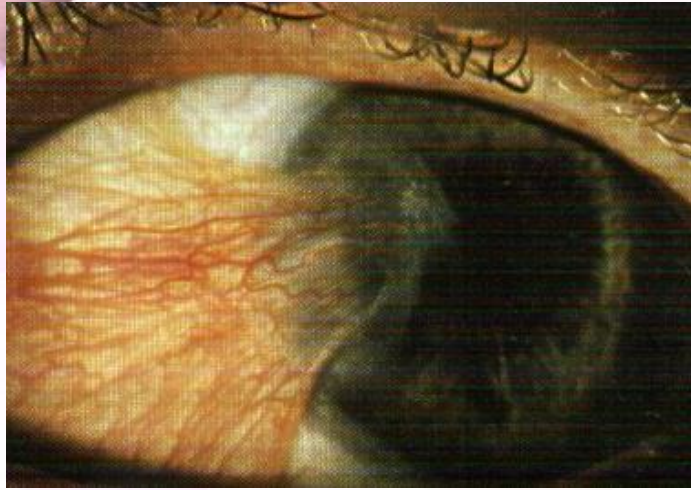
- Simtomatis dan suportif
- Resolusi spontan dlm 2 minggu
- **Hindari** → STEROID

TRAKHOMA

- personal hygiene
- tetracycline, 1 – 1.5 gr/d , orally –divided in 4 doses, 3 –4 weeks
- doxycycline, 2 x 100 mg, 3 weeks
- topical : sulfonamide, tetracycline, erythromycin
- surgical correction (trichiasis)

DEGENERATIVE PROCESS

PTERYGIUM



- TRIANGULAR FIBROVASCULAR SUBEPITHELIAL INGROWTH OF DEGENERATIVE BULBAR CONJUNCTIVAL TISSUE OVER THE LIMBUS ONTO THE CORNEA
- Hot climate, response to chronic dryness and uv exposure

PTERYGIUM

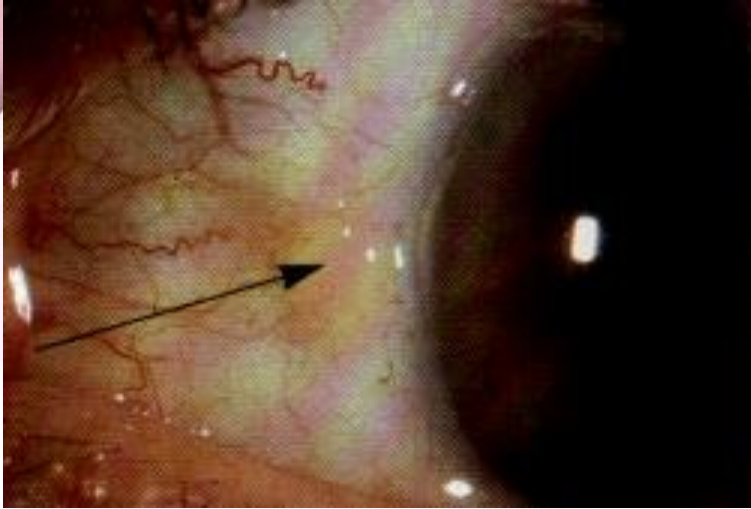
COMPLICATIONS

- Cosmetic
- Chronic irritation
- Decrease vision due to visual axis involvement / astigmatism
- Disruption of the precorneal tear film
- Inflammation

TREATMENT

- Simple excision → high recurrence
- Excision with conjunctival autograft / amniotic membrane
- Adjunctive treatment → mitomycin C, beta irradiation

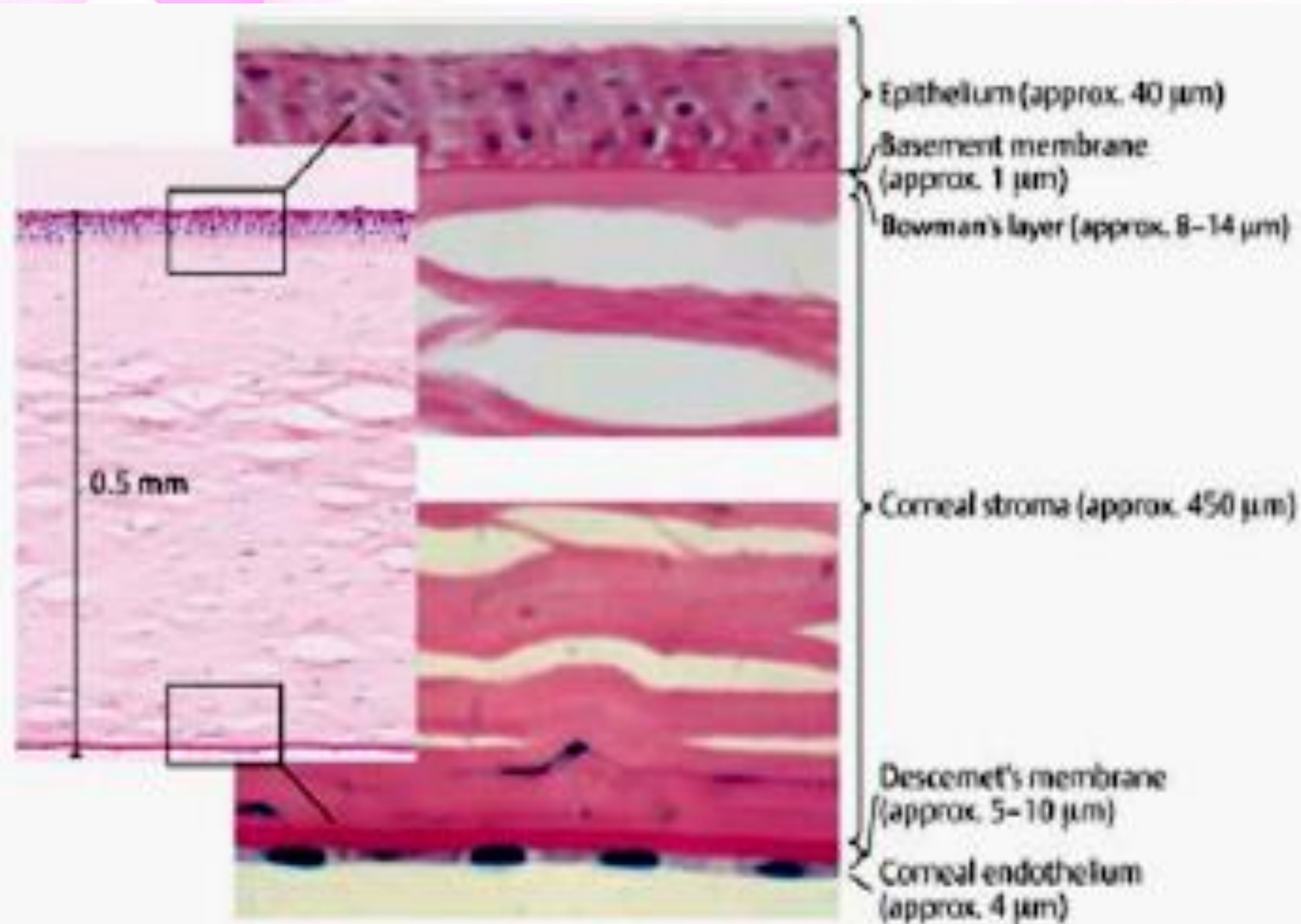
PINGUECULA



- ❑ YELLOW-WHITE DEPOSITS ON THE BULBAR CONJUNCTIVA ADJACENT TO THE NASAL OR TEMPORAL LIMBUS
- ❑ TREATMENT → if inflammed, use steroid

CORNEA

- Avaskuler, transparan
- Nutrisi :
 - difusi glukosa dari COA
 - O₂ dari tear film
 - kornea perifer → O₂ dari sirkulasi limbal
- Sensori nerve fiber : N.ciliaris longus → plexus sub epitel → cab. N.V → multiple, unmyelinated → nyeri
- Sel stem epitelial di superior dan inferior limbus



LAPISAN KORNEA

1. Epitel

- Sel epitel skuamosa stratified, non keratinisasi
- Epitel dan tear film → optical smooth surface
- Tight junction antara sel epitel superfisial mencegah penetrasi airmata masuk stroma

2. Membrana Bowman

- Lap. aseluler jernih
- Fibril kolagen
- terjadi skar bila rusak

3. Stroma

- 90 % tebal kornea
 - T/D - serabut kolagen
 - sel keratocyt
- } dalam bahan dasar Mukopolisakarida
- Serabut kolagen → paralel teratur → transparan

4. Membrana Descemet

- Lap. Terkuat ; tidak mudah ditembus
- Melapisi stroma di bagian posterior
- terdiri dari serat kolagen jernih
- Dianggap sebagai hasil sekresi endotel

5. Endotel

- T/D 1 lapis sel kubus
- Regenerasi (-), pompa bikarbonat aktif
- Kerusakan permanen → jumlah menurun sejalan dg usia

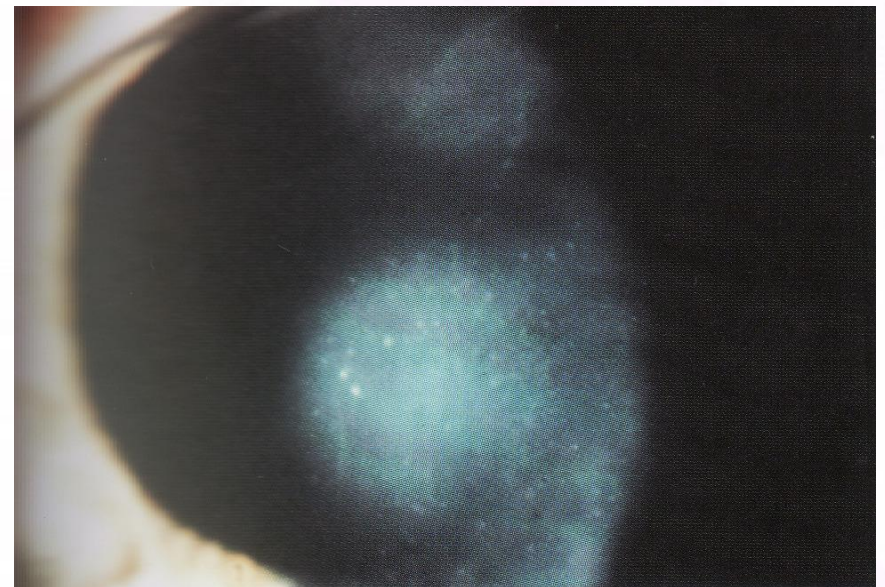
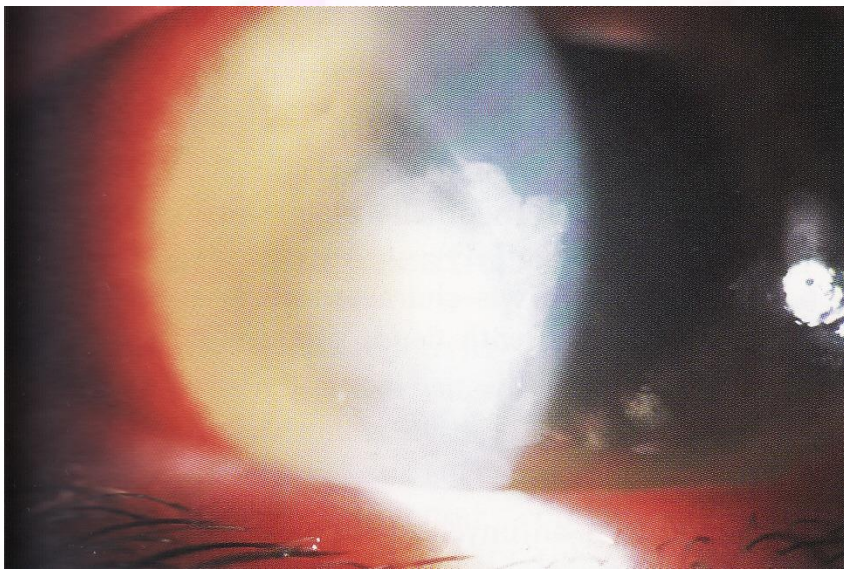
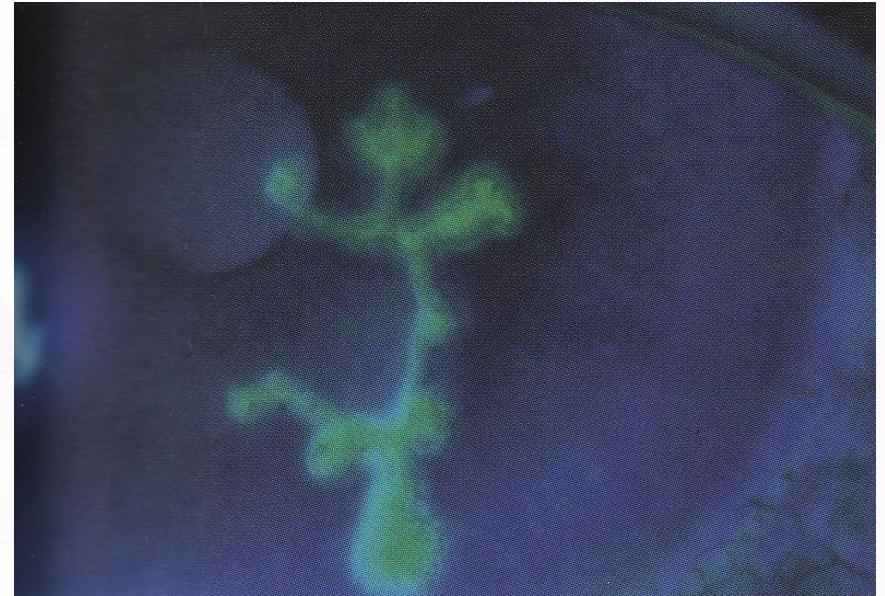
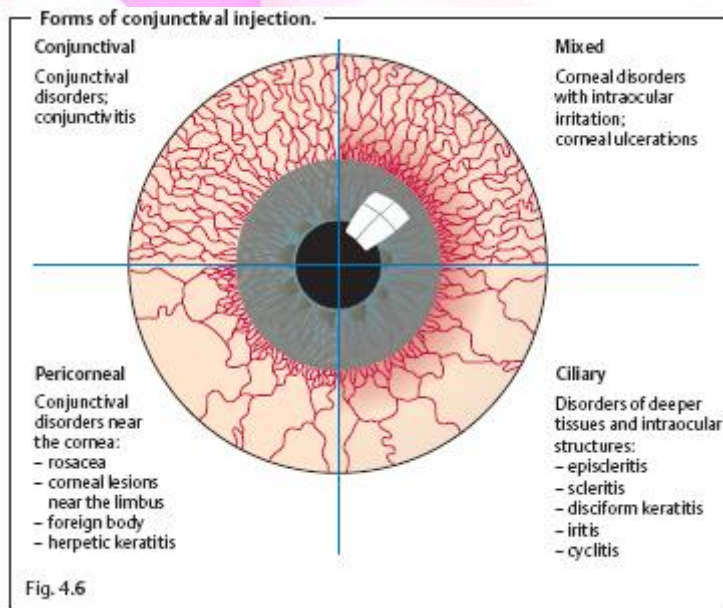
KERATITIS

GEJALA

- KABUR
- SILAU
- MATA MERAH
- NYERI

TANDA

- BLEFAROSPASME
- INJEKSI PERIKORNEA
- INFILTRAT KORNEA
 - Distribusi : Difus, Fokal, Multifokal
 - Kedalaman : Epitel, Sub Epitel, Stromal, Endotel
 - Lokasi : Sentral, Perifer
 - Bentuk : Dendrit, Disciform, Numular DII



PREDISPOSISI FAKTOR

- BLEFARITIS
- LENSA KONTAK
- LAGOFTALMUS
- TRAUMA
- OBAT-OBAT IMUNOSUPRESIF
TOPIKAL/SISTEMIK
- DLL

PATOGEN PENYEBAB

- BAKTERI
- VIRUS
- ACANTHAMOEBA
- FUNGI

KERATITIS BAKTERIAL

> 90% KERADANGAN KORNEA O.K BAKTERI

Visus ↓, Fotofobi, Sakit, Nrocoh, Merah, Blefarospasme,
Infiltrat

Penyebab:

- Staphylococcus aureus
- Staphylococcus epidermidis
- Streptococcus pneumonia dan Strep. Sp.
- Pseudomonas aeruginosa
- Enterobacteriaceae

Diagnosis etiologi :

1. Klinis
2. Laboratoris

Kerokan kornea dan pemeriksaan mikrobiologi gram, kultur, uji resistensi

Bacterium	Typical characteristics of infection
<i>Staphylococcus aureus</i>	Infection progresses slowly with little pain.
<i>Staphylococcus epidermidis</i>	As in <i>Staphylococcus aureus</i> infection.
<i>Streptococcus pneumoniae</i>	Typical serpiginous corneal ulcer; the cornea is rapidly perforated with early intraocular involvement; very painful.
<i>Pseudomonas aeruginosa</i>	Bluish green mucoid exudate, occasionally with a ring-shaped corneal abscess. Progression is rapid with a tendency toward melting of the cornea over a wide area; painful.
<i>Moraxella</i>	Painless oval ulcer in the inferior cornea that progresses slowly with slight irritation of the anterior chamber.

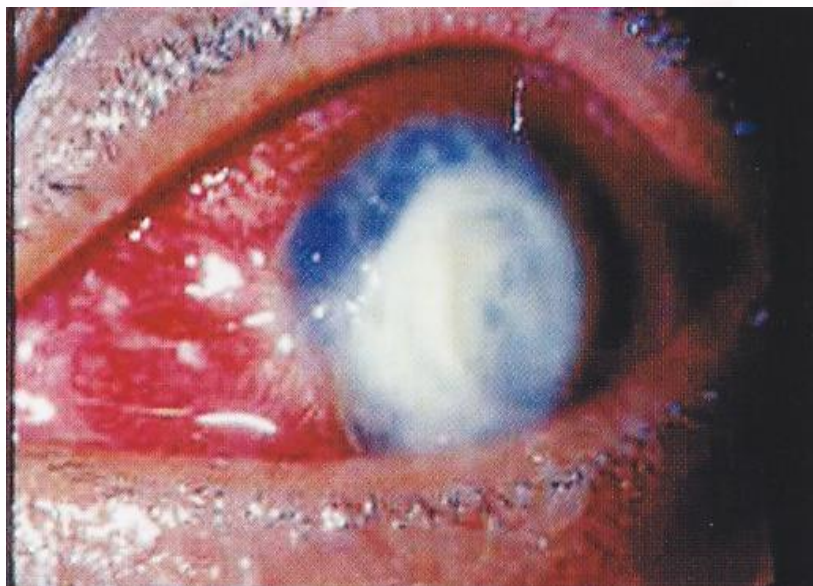
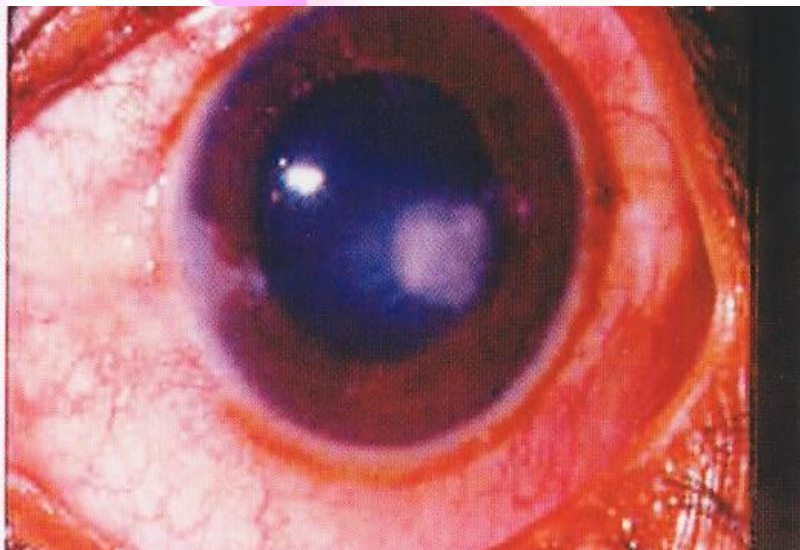
Terapi

INITIAL THERAPY FOR BACTERIAL KERATITIS

ORGANISM	ANTIBIOTIC	TOPICAL DOSE	SUBCONJUNCTIVAL DOSE
Gram-positive cocci	Cefazolin	50 mg/mL	100 mg in 0.5 mL
	Vancomycin*	50 mg/mL	25 mg in 0.5 mL
Gram-negative rods	Tobramycin	9–14 mg/mL	20 mg in 0.5 mL
	Ceftazidime	50 mg/mL	100 mg in 0.5 mL
	Fluoroquinolones	3 mg/mL	Not available
No organism or multiple types of organisms	Cefazolin with Tobramycin	50 mg/mL	100 mg in 0.5 mL
	or	9–14 mg/mL	20 mg in 0.5 mL
	Fluoroquinolones	3 mg/mL	Not available
Gram-negative cocci	Ceftriaxone	50 mg/mL	100 mg in 0.5 mL
	Ceftazidime	50 mg/mL	
Mycobacteria	Amikacin	20 mg/mL	20 mg in 0.5 mL

Obat Penunjang

- sikloplegi, analgesik, anti glaukoma



KERATITIS JAMUR

Gx Kx : sakit kepala, reaksi radang tidak berat dan tidak akut
Rx trauma / tumbuh-tumbuhan
hifa, infiltrat satelit, plaque endotel,
hipopion dan membran, gray white, filamentous.

Terapi

Natamycin 5%, Amphotericin B (0.15 – 0.3 %)
Itraconazole oral 200 mg/hari

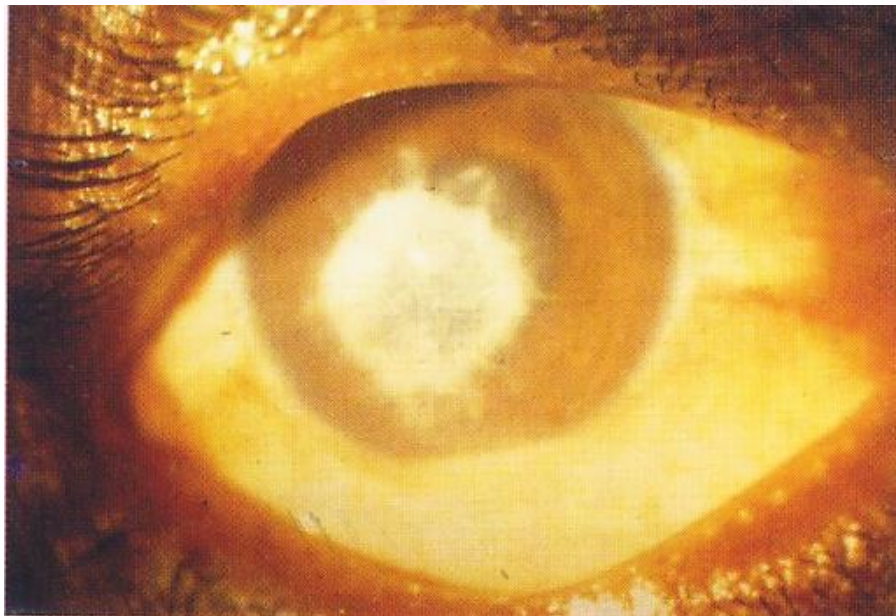
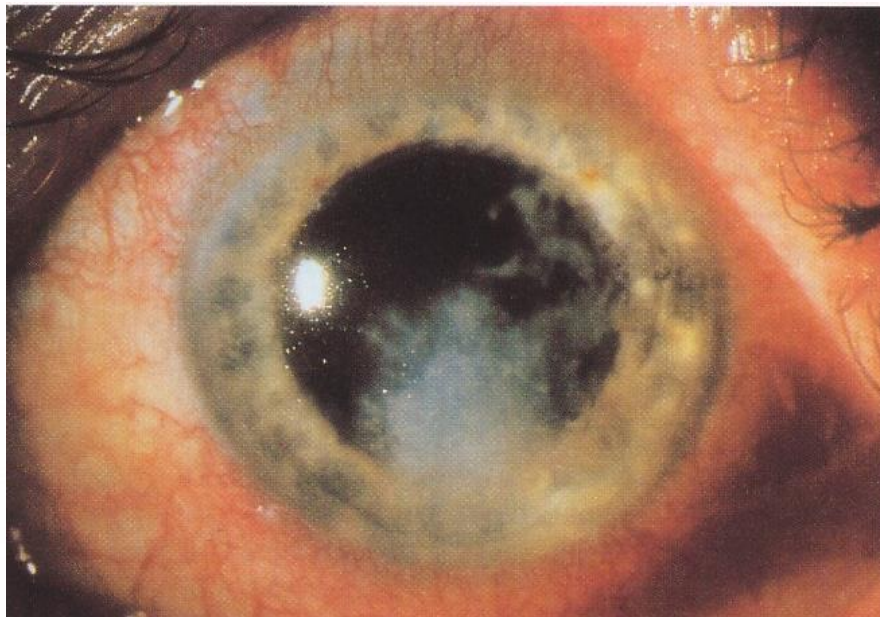


Table-1: Typical clinical features

Features of bacterial ulcer	Features of fungal ulcer
1. History of trauma to the cornea, contact lens wear 2. Pain, redness, watering, decrease in vision 3. Lid oedema (marked in gonococcal ulcer), purulent discharge in gonococcal ulcer and bluish green discharge in pseudomonas corneal ulcer 4. Round or oval in shape involving central or para central part of the cornea. Rest of the cornea is clear. Hypopyon may or may not be present. 5. In pneumococcal ulcer the advancing border will have active infiltrate with undermined edges and the trailing edge may show signs of healing. Most of the pneumococcal ulcers will show leveled hypopyon associated with Dacryocystitis. 6. Pseudomonas ulcer will have short duration, marked stromal oedema adjacent to the ulcer with rapid progression. If untreated, will perforate within 2-3 days. Advanced ulcer may involve the sclera also. 7. Ulcers caused by Moraxella and Nocardia are slowly progressive in immunocompromised hosts.	1. History of trauma with vegetable matter 2. Suspect fungal ulcer if patient reports agriculture as main occupation. 3. Pain and redness are similar to bacterial ulcer. But lid oedema is minimal even in severe cases unless patients have received native medicines or peri ocular injections. 4. Early fungal ulcer may appear like a dendritic ulcer of herpes simplex virus. The feathery borders are pathognomonic clinical features. Satellite lesions, immune ring, and unlevelled hypopyon may aid in diagnosis. 5. The surface is raised with greyish white creamy infiltrates, which may or may not appear dry. 6. Ulcer due to pigmented fungi will appear as brown or dark; raised, dry, rough, leathery plaque on the surface of the cornea

KERATITIS VIRUS HERPES SIMPLEKS

Patogenesis :

- infeksi HSV :
- kontak langsung
 - paparan virus
 - infeksi genital

infeksi primer HSV I : dari infeksi kulit dan epitel mukosa
via akson saraf sensori
→ inf.laten (pada ganglion saraf
sensori trigeminal)

Gx (infeksi okuler primer)

- unilateral blefarokonjungtivitis
- radang konjungt → follicular
- keratitis epitel

INFEKSI OKULER REKUREN

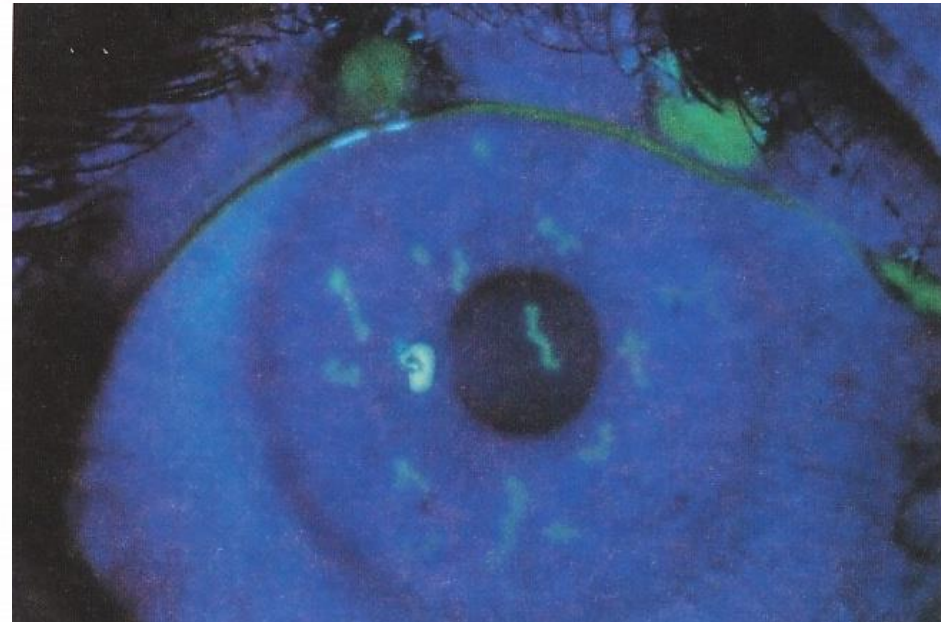
Patogenesis : - reaktivasi virus

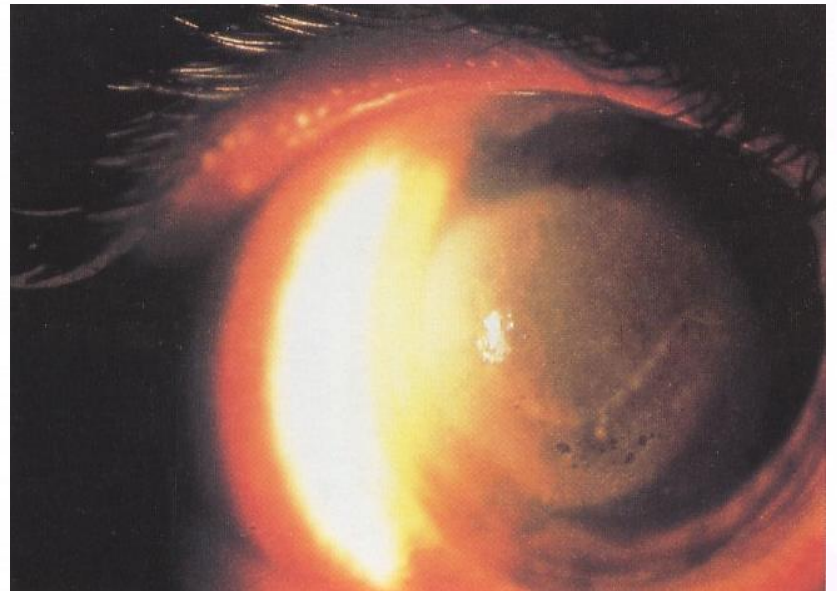
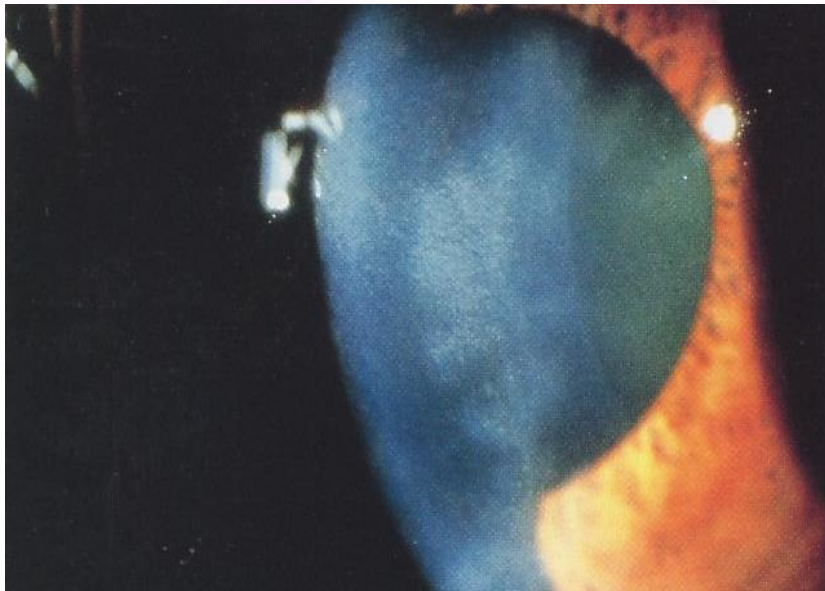
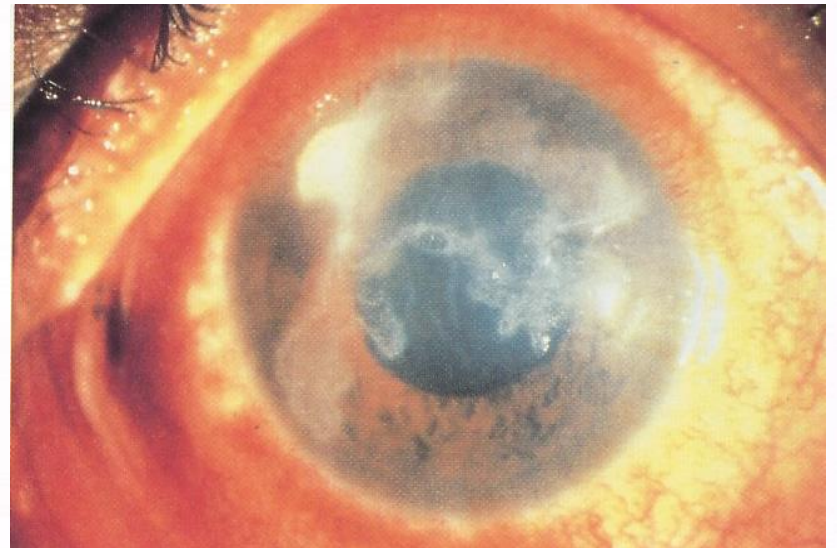
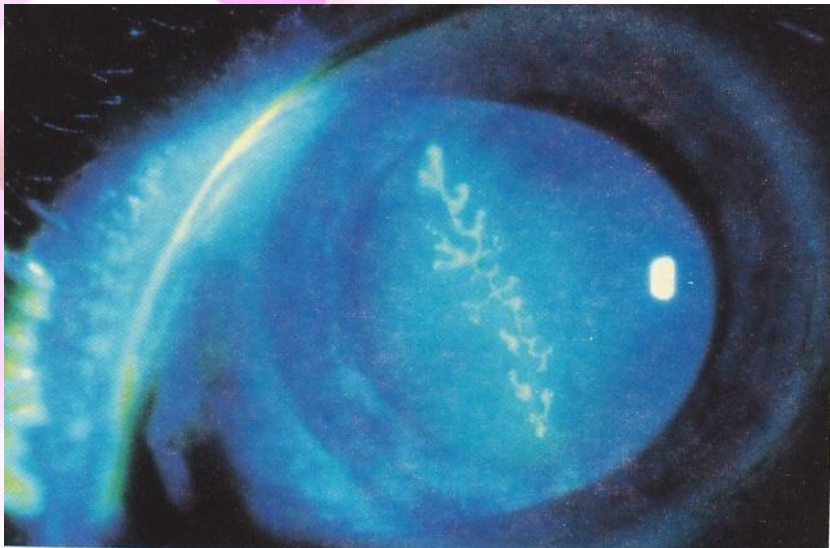
- transportasi virus ke axon saraf

→ nerve ending sensorik → permukaan epitel

Gx Kx :

- blefarokonjungtifitis
- keratitis epitel dan stroma
- iridosiklitis





ANTIVIRAL AGENTS IN EXTERNAL/CORNEAL INFECTIONS WITH HERPES SIMPLEX VIRUS

AGENT	MECHANISM OF ACTION	ADMINISTRATION	DOSAGE FOR ACUTE DISEASE
Vidarabine	Purine analogue Inhibits DNA polymerase	3% ophthalmic ointment ¹	5×/day for 10 days
Trifluridine	Pyrimidine analogue Blocks DNA synthesis	1% ophthalmic solution	8×/day for 10 days
Acyclovir	Activated by HSV thymidine kinase to inhibit viral DNA polymerase	3% ophthalmic ointment ² 200, 400, 800 mg; 200 mg/5 mL suspension 5% dermatologic ointment ³	5×/day for 10 days 400 mg 5×/day for 10 days 6×/day for 7 days
Famciclovir ⁴	Pro-drug of penciclovir	125, 250, 500 mg	250 mg 3×/day for 10 days
Valacyclovir ⁴	L-valyl ester of acyclovir	500, 1000 mg	1000 mg 2×/day for 10 days
Penciclovir	Inhibits viral DNA polymerase	1% dermatologic cream ³	8×/day for 4 days

¹No longer manufactured; can be obtained through compounding pharmacies.

²Not commercially available in the United States.

³Not for ophthalmic use.

⁴Optimal dose for ocular disease not determined.

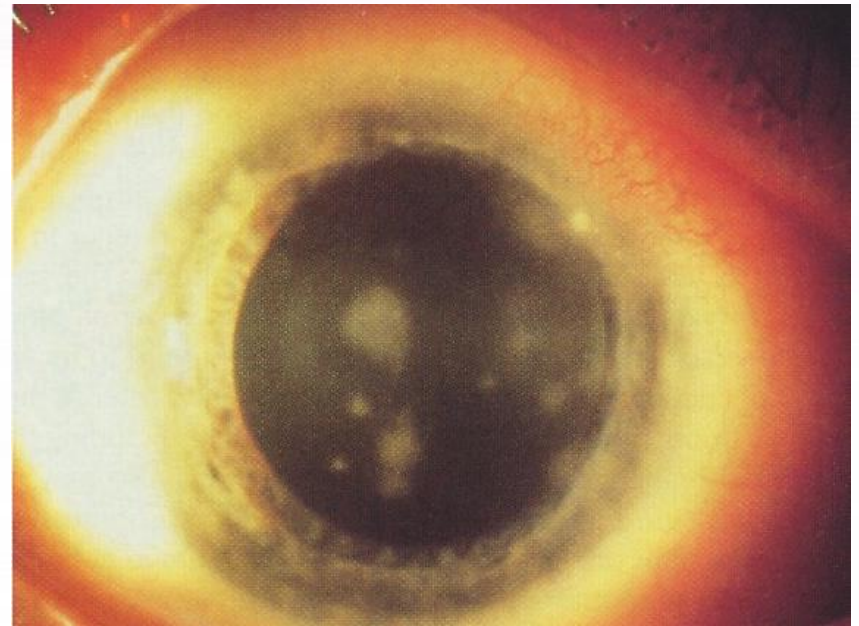
KERATITIS HERPES ZOSTER OFTALMICUS

Patogenesis : laten di sensori nerve, reaktivasi endogen

Gx :

- maculo papular rash, vesikel panas, merah
- neuralgia
- N.V cabang ophtalmika
- keratitis epitel dendrit, epitel pungtat
- sensasi kornea turun
- keratitis stromal : numular, interstitial, disciform
- uveitis anterior
- orbita – CNS : arteritis oklusif → ptosis, proptosis, edem orbita

Terapi : - acyclovir 800 mg 5 x 1 (7 – 10 hr)
- Corticosteroid, sikloplegik

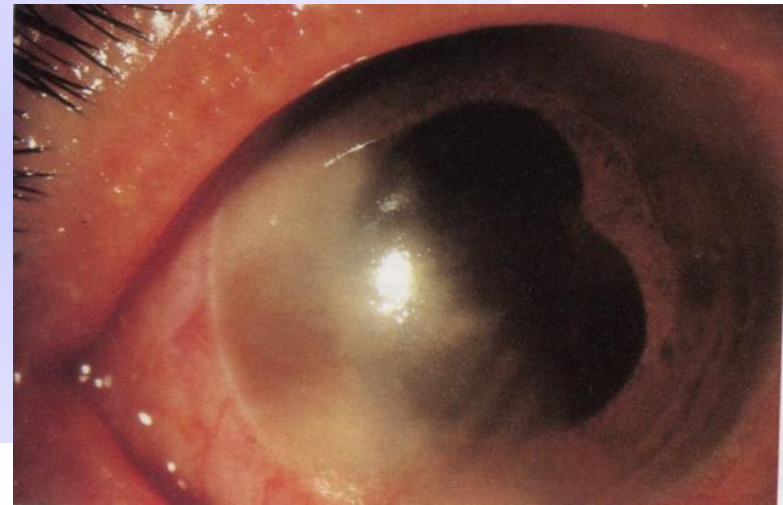


IMMUN MEDIATED DISEASE OF THE CORNEA

KERATITIS INTERSTITIAL

Patogenesis :

- infiltrasi sel dan vaskularisasi pada stroma kornea tanpa mengenai epitel dan endotel
- respon hipersensitivitas IV
- Berhub. dengan penyakit infeksius (TBC, lepra, rubeola, dll)



INFILTRAT KORNEA MARGINAL

Patogenesis :

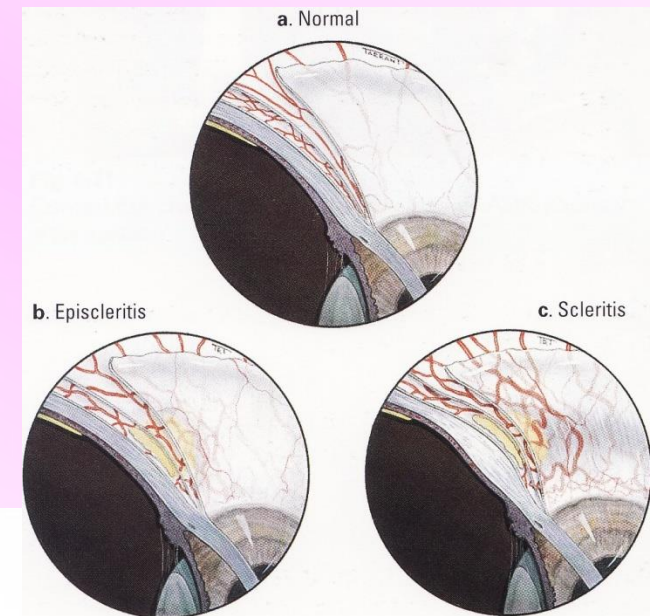
- Limbus → peran pada imun mediated kornea
- Kornea perifer berdekatan dengan vaskularisasi limbus
- Predisposisi : blefarokonjungtivitis, trauma, lensa kontak, dll

Gx Kx :

- infiltrat → posisi jam 10, 2, 4, 8
- pada blefaritis stafilokokal zona interval : positif dari limbus
- penyakit kronis : pembuluh darah superfisial pada area infiltrat

SCLERA

- Anterior episklera → jaringan penghubung bervaskularisasi, diantara stroma sklera superfisial dan capsula Tenon.
- Stroma sclera → jaringan kolagen
- Bagian dalam sclera (lamina fusca) → bergabung dg lamela supracilliary dan suprachoroid tractus uvea
- Lapisan vaskularisasi :
 - conjunctival vessel
 - Tenon capsule
 - deep vascular plexus



IMMUN MEDIATED DISEASE OF THE SCLERA AND EPISCLERA

EPISCLERITIS

Patogenesis : self limited, tidak jelas, sistemik

Gx Kx :

- transien, 20 – 50 th, pada area pinguikula
- tanda radang terlokalisir pada episklera
- warna merah terang, salmon pink
- dengan penileprin topikal → hilang

Klasifikasi :

- simpel (injeksi difusa) dan nodular



Terapi :

- work up → underlying causes
- tanpa terapi membaik
- NSAID topikal atau oral, lubrikan
- bila tidak respon → kortikosteroid

SCLERITIS

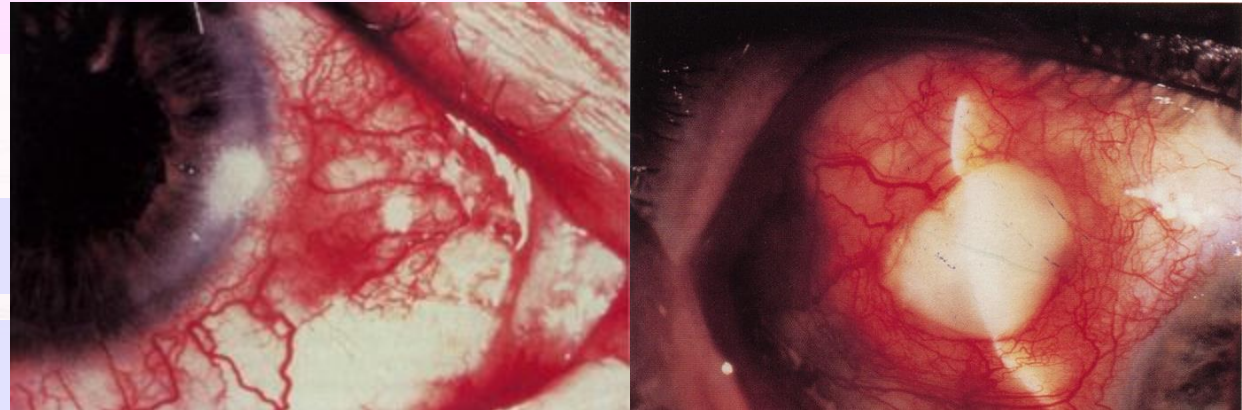
Patogenesis : - imun mediated → vaskulitis
- underlying sistemic immunologic disease

Gx Kx :

- dekade IV – VI, wanita, sakit kepala pada sisi sakit
- warna : violaceous hue
- p.d. sklera crisscross pattern, melekat ke sklera
- dengan aplikator cotton – tipped → tidak bergerak

Skleritis Anterior Noduler

- Nodul sklera, immobil dan deep red purple, lepas dari jaringan episklera



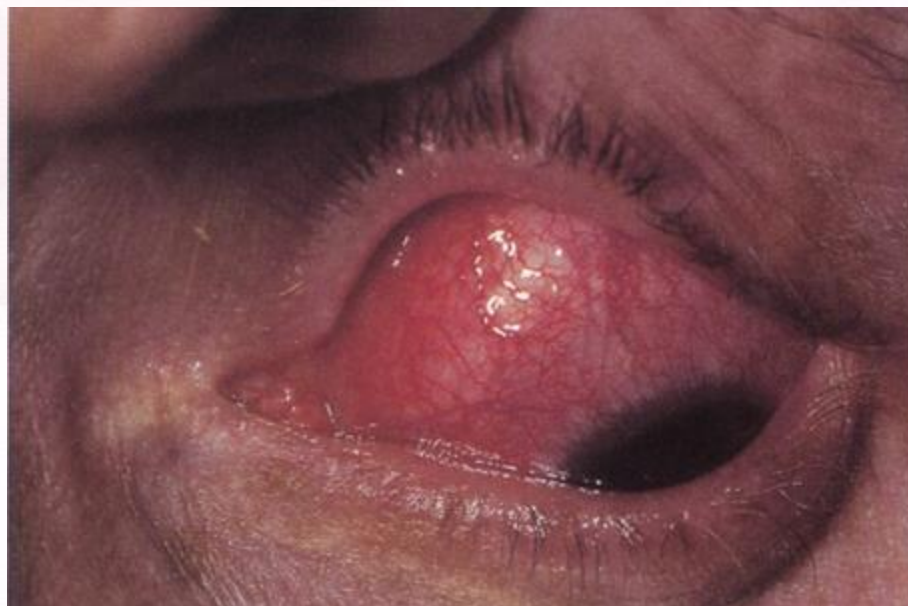
Necrotizing Skleritis

* Dengan inflamasi

- Sakit >>>, lokal inflamasi, edem avaskuler
- Sklera blue gray appearance (akibat thinning)

* Tanpa inflamasi (sclero malacia perforan)

- R.A. tanda radang minimal, tidak sakit, penipisan sklera
- Tampak jar.Uvea, Bulging stafiloma



Komplikasi skleritis

- Keratitis perifer, uveitis, katarak, glaukoma, sklera thinning

Lab → penyakit sistemik

- Konsul internis, rheumatologi

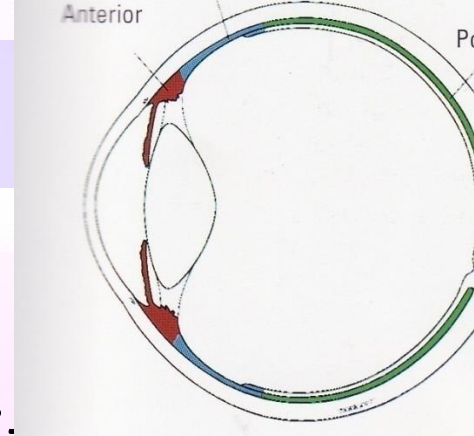
Terapi :

- Kortikosteroid topikal, oral, IV
- NSAID oral
- Imunosupresif

UVEITIS

KLASIFIKASI ANATOMIS

1. Anterior uveitis → iritis, iridocyclitis
2. Intermediate uveitis → s.d pars plana, retina perifer, dan koroid dibawahnya
3. Posterior uveitis → retina posterior dan koroid + vitreus
4. Panuveitis → seluruh tractus uvea



KLASIFIKASI KLINIS

- Akut → sudden, symptomatic onset, 3 mo.
- Kronis → > 3 mo, insidious / asymptomatic

KLASIFIKASI ETIOLOGIS

1. Berhubungan dg penyakit sistemik
2. Infeksi bakteri dan virus → TB, herpes zoster
3. Infestasi protozoa atau nematoda
4. Idiopathic specific uveitis → Fuchs uveitis syndrome
5. Idiopathic non-specific uveitis

UVEITIS ANTERIOR

GEJALA

- AKUT → fotofobia, nyeri, merah, VA turun, lakrimasi
- KRONIS → asimtomatis, floater

TANDA

1. Injeksi silier

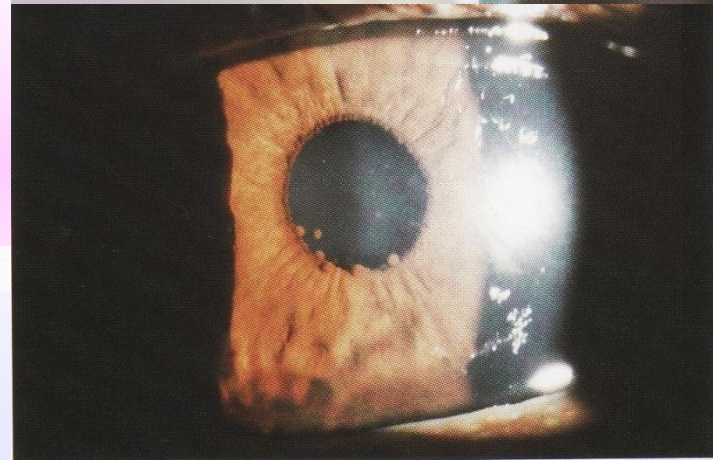


TANDA

2. Keratic presipitat (KP)

3. Sel dan flare

4. Iris nodule



KOMPLIKASI

- Sinekia posterior
- Lain-lain → band keratopathy, katarak, glaukoma, edema makula, cyclitic membran formation, phtisis bulbi

TERAPI

- Midriatikum → sikloplegik
- Steroid → topikal, injeksi periokular, injeksi intravitreal, sistemik
- Agen imunosupresif → antimetabolit, T-cell inhibitor



SELAMAT BELAJAR